Ericsson PE Node YEL-PE-11 SOP: Resolving QoS / Congestion Event on Major Alarm ALARM_3572

Node: YEL-PE-11 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_3572

Description

This Standard Operating Procedure (SOP) is designed to address the specific failure domain of QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on Ericsson PE Node YEL-PE-11 at Yelahanka Sub-station. The underlying protocol mechanics responsible for this failure domain involve the interaction of the QoS (Quality of Service) mechanism, which is implemented to prioritize traffic based on pre-defined classes of service, and the congestion control mechanism, which is designed to prevent network overload by policing and shaping traffic. In this specific scenario, the severity of the failure is classified as Major, indicating a significant impact on network performance. The root cause of this failure is likely to be related to misconfiguration, hardware or software issues, or high-traffic volume. It is essential to execute this SOP with caution, as incorrect execution may exacerbate the issue or introduce new problems.

Pre-requisite Checks

- Confirm the Node ID: YEL-PE-11 is accessible remotely or physically.

- Verify the Site Location: Yelahanka Sub-station is in a stable condition.

- Check the redundant link status on the Ericsson PE Node YEL-PE-11.

- Ensure the CLI user has the appropriate privileges and access rights to execute diagnostic and resolution commands.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies or parameters	High
Hardware issues with the interface or line card	Medium
Software issues or bugs in the Ericsson PE Node YEL-PE-11 firmware	Low
High-traffic volume or bursty traffic patterns	Medium

Diagnostic Commands & Expected Outputs

```
Command: show qos policy, Expected Output: Policy name, priority, class-of-service, and policing parameters, Description: This command is used to verify the QoS policy configuration on the Ericsson PE Node YEL-PE-11, including the priority, class-of-service, and policing parameters. The expected output will indicate if the QoS policy is correctly configured.
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```
Command: show interface, Expected Output: Interface status, traffic counters, and error statistics, Description: This command is used to verify the interface status, traffic counters, and error statistics on the Ericsson PE Node YEL-PE-11. The expected output will indicate if there are any issues with the interface or line card.

Command: show process cpu, Expected Output: CPU usage and load average, Description: This command is used to verify the CPU usage and load average on the Ericsson PE Node YEL-PE-11. High CPU usage may indicate software issues or bugs in the firmware.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: configure qos policy , Description: This step involves verifying and correcting the QoS policy configuration on the Ericsson PE Node YEL-PE-11. The correct configuration will ensure that traffic is properly prioritized and policed.

Step 2: Step: Reset or replace faulty interface or line card, Command: reset interface , Description: This step involves resetting or replacing the faulty interface or line card on the Ericsson PE Node YEL-PE-11. This will ensure that the interface or line card is functioning correctly and not contributing to the congestion issue.

Step 3: Step: Update or patch firmware to resolve software issues, Command: download software , Description: This step involves updating or patching the firmware on the Ericsson PE Node YEL-PE-11 to resolve software issues or bugs. This will ensure that the firmware is up-to-date and functioning correctly.

Post-Check Verification

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Command: show qos policy, Expected Output: Policy name, priority, class-of-service, and policing parameters, Description: This command is used to verify that the QoS policy configuration on the Ericsson PE Node YEL-PE-11 has been correctly modified or updated.

Command: show interface, Expected Output: Interface status, traffic counters, and error statistics, Description: This command is used to verify that the interface status, traffic counters, and error statistics on the Ericsson PE Node YEL-PE-11 have been corrected or updated.

Command: show process cpu, Expected Output: CPU usage and load average, Description: This command is used to verify that the CPU usage and load average on the Ericsson PE Node YEL-PE-11 have been corrected or updated.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: configure qos policy , Description: This step involves restoring the original QoS policy configuration on the Ericsson PE Node YEL-PE-11 in case the fix fails.

- Step: Restore original interface or line card configuration, Command: configure interface , Description: This step involves restoring the original interface or line card configuration on the Ericsson PE Node YEL-PE-11 in case the fix fails.

- Step: Restore original firmware version, Command: download software , Description: This step involves restoring the original firmware version on the Ericsson PE Node YEL-PE-11 in case the fix fails.